MIL-T-21205B(WEP)
15 November 1960
Superseding
MIL-T-21205A(AER)
24 June 1958

MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, FUEL; TRU-12/A (300 - 5000 - 15000 PPH)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers the design and performance requirements for the procurement of one type of a remote indicating, fuel, rate of flow transmitter.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

P-S-661 - Solvent, Dry - Cleaning

Military

MIL-P-116 - Preservation, Method of
MIL-H-3136 - Standard Test Fluids, Hydrocarbon and Iso-octane
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5624 - Jet Fuel, Grades JP-3, JP-4, and JP-5
MIL-F-7024 - Fluids, Calibrating, For Aircraft Fuels, Systems Components
MIL-I-7566 - Indicator System, Fuel Flowmeter, General Specification for
MIL-P-7936 - Parts and Equipments, Aeronautical, Preparation for Delivery
MIL-I-21206 - Indicator, Rate of Flow, Fuel; EFU-7/A (300-5000-15,000 PPH)
MIL-C-26482 - Connectors, Electric, Circular, Miniature, Quick Disconnect

FSC 6620

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STANDARDS

Military

MIL-STD-704 - Electric Power, Aircraft, Characteristics and Utilization of
MS3134 - Connector, Receptacle, Electric, Single Hole Mounting, Push-Pull Type

Air-Force - Navy Aeronautical Drawings

AND10050 - Bosses, Standard Dimensions for Gasket Seal Straight Thread

(When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention: Code CDS.)

3. REQUIREMENTS

3.1 Qualification. - The transmitter furnished under this specification shall be a product which has been tested and has passed the qualification tests herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General specification. - The requirements of Specification MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction . -

3.3.1 General. - The transmitter shall be designed and calibrated to operate satisfactorily with a 300-5000-15000 pph indicator in accordance with Specification MIL-I-21206.

3.3.1.1 The transmitter remote indication system shall operate as a synchro system in which the moving element consists of a 2-pole wound rotor, designed for operation with 115V 400 cycle single phase ac. The stator is 3 phase Y-connected.

3.3.2 Range. - The transmitter shall measure flow rated from 300-5000-15000 pounds per hour.

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3.3.3 Operating pressure. - The high pressure transmitter shall operate under a pressure of 600 psi and withstand a pressure of 1600 psi without damage.

3.3.4 Pressure drop. - The transmitter, by virtue of its design, shall not have appreciably more pressure drop under any fuel flow conditions when inoperative than when it is operating normally.

3.3.5 Temperature range. - The transmitter shall be designed to withstand an ambient operating temperature range of -55° to +121°C and high temperature exposure to +150°C.

3.3.6 Dimensions. - The outline dimensions of the transmitter shall conform to Fig. 1.

3.3.7 Mounting. - Mounting bosses and spacing thereof shall be as specified on Fig. 1.

3.3.8 Flow direction. - The direction of fuel flow through the transmitter shall be permanently and legibly marked by arrow as shown on Fig. 1 on both front and back.

3.3.9 Weight. - The weight of the complete fuel flowmeter transmitter shall not exceed 6.5 pounds.

3.4 Electrical. -

3.4.1 Power requirement. -

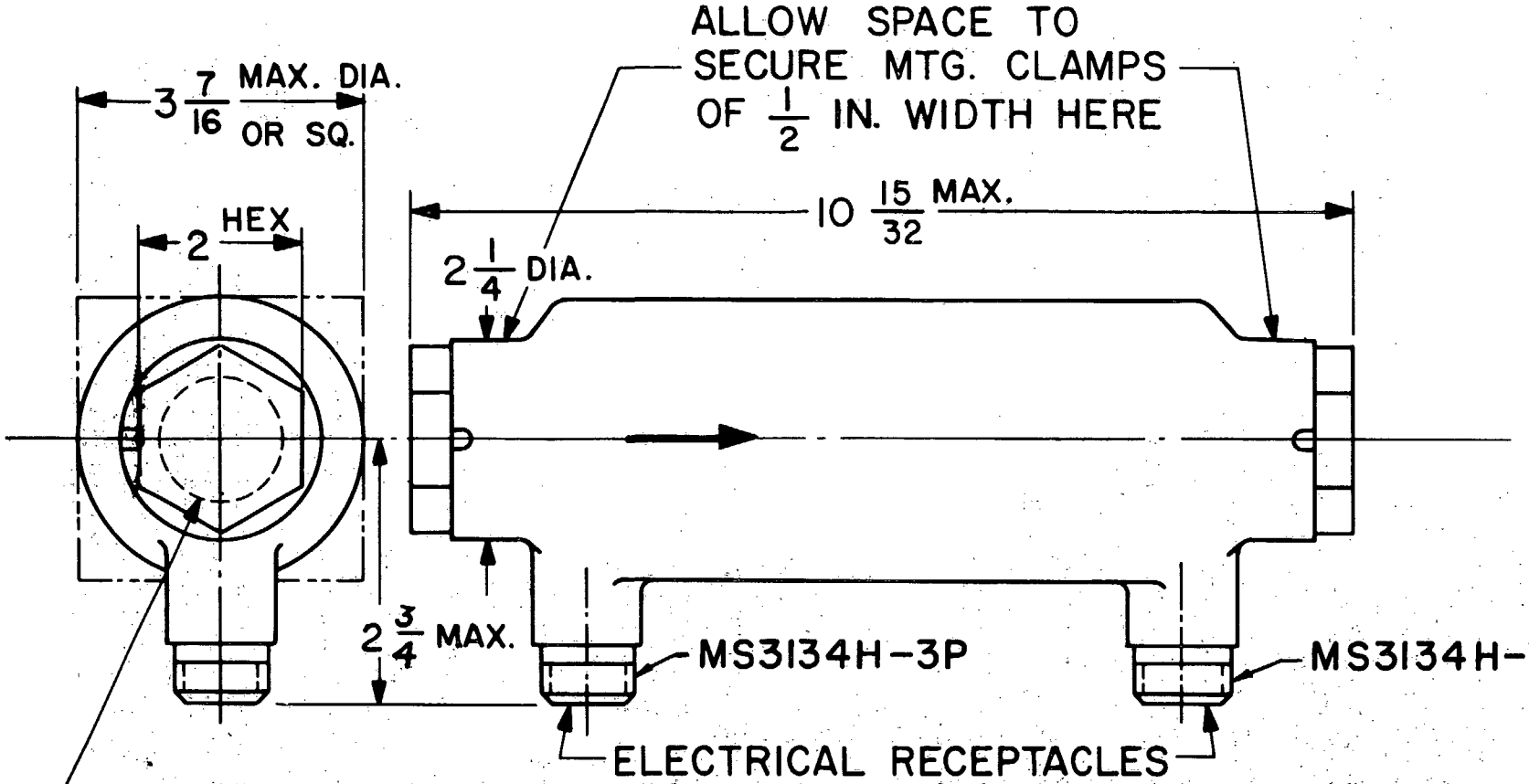
3.4.1.1 Synchro system. - The synchro system of the transmitter shall be designed to operate from 115V ±20 percent, 400 cps ±20 percent single phase ac power and as otherwise specified in Standard MIL-STD-704, except that the ratio of volts to frequency shall be 115/400 ±5 percent.

3.4.1.2 Motor. - The motor of the transmitter shall be designed to operate from 115V ± 30 percent, 400 cps ±0.25 percent.

3.4.2 Power consumption. - The power consumption of the transmitter shall be a maximum of 15 watts.

3.4.3 Undervoltage protection. - The transmitter shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within its correct range.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS $\pm \frac{1}{64}$

TRANSMITTER
FIGURE 1.

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3.4.4 Wiring. - The external wiring of the transmitter shall be as shown in Figure 2. Internal wiring shall be neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be tied to the terminals prior to soldering, where practicable. The wiring shall be insulated from the transmitter housing.

3.4.5 Electrical receptacle. - The transmitter shall be provided with two electrical receptacles conforming to Specification MIL-C-26482 as shown in Figure 1. One receptacle shall be Standard MS3134H-3P and the other receptacle shall be Standard MS3134H-7P.

3.5 Performance. - The transmitter shall perform satisfactorily when subjected to the tests specified in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - Sampling, inspection and testing of the transmitters shall be as specified in Specification MIL-I-7566, and as specified herein.

4.1.1 Applicable tolerances shall be as specified in Section 3, and as specified in Tables I, II, and III.

4.1.2 Test position. - Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.1.3 Test fluid and fuel. - The fluid used for calibrating and testing the transmitter shall be fluid conforming to Specifications P-S-661, MIL-H-3136 Type I or MIL-F-7024 Type II. Aviation fuel conforming to Specification MIL-F-5624 Grades JP-4 and JP-5 may be used.

4.1.4 Fuel temperature. - The temperature of the test fuel during high temperature test runs shall be 38 ±2°C. The fuel shall be pumped through the transmitter at the rate of 1000 pph for the last 30 minutes prior to scale and friction error calibration tests.

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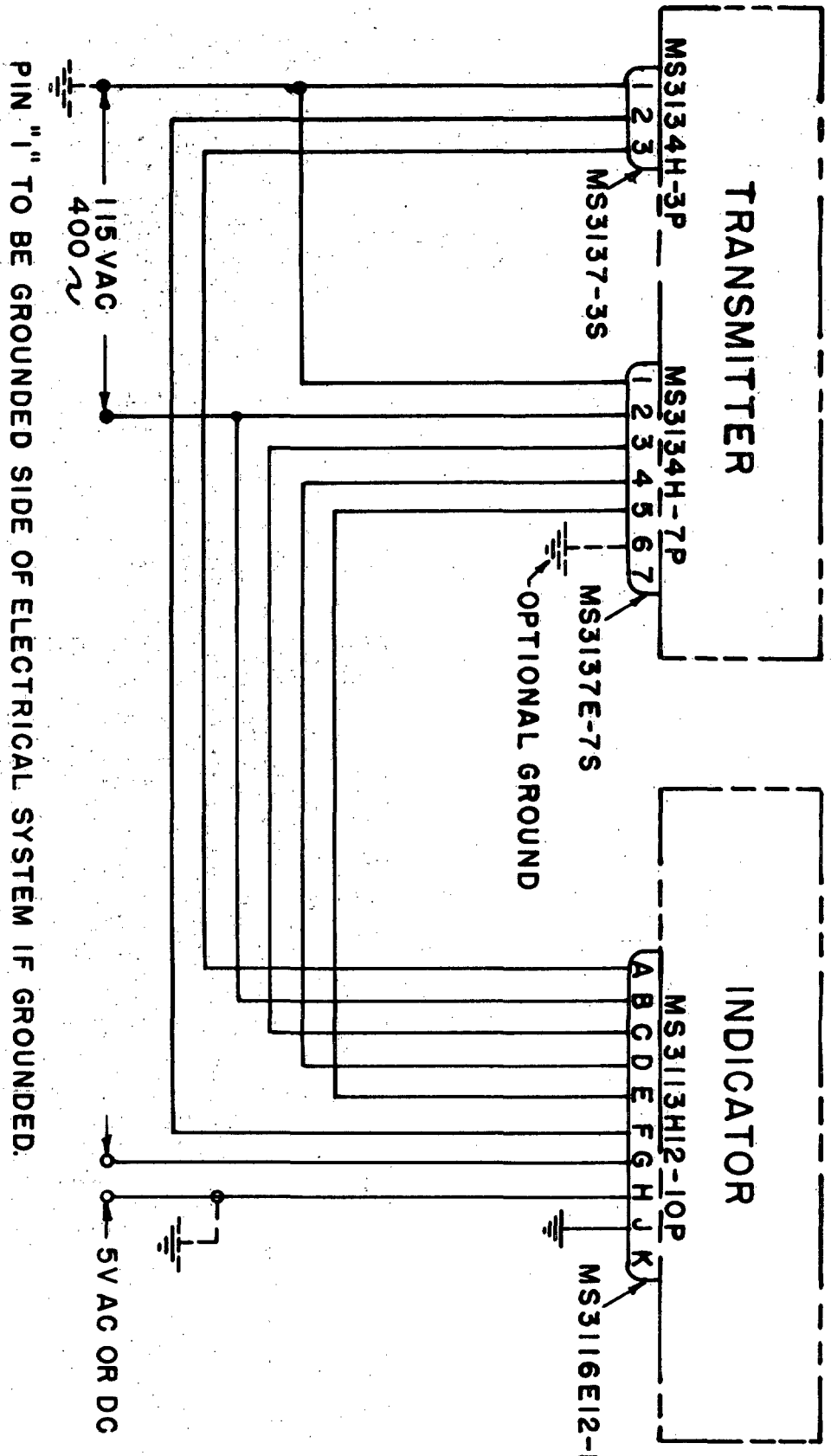


FIG. 2. EXTERNAL WIRING DIAGRAM

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TABLE I

FUEL FLOWMETER TRANSMITTER

Flow Rate (PPH)	Scale Error Tolerance (±PPH) Room Temp.	High Temp.	Low Temp.
300	45	55	75
500	35	45	65
1500	35	45	60
2500	30	40	55
3500	30	40	55
4500	45	60	85
6000	120	240	300
9000	170	340	425
12000	210	420	510
15000	250	500	625

TABLE II

Test	Tolerance PPH ± 500-4900 PPH Scale	5000-15000 PPH Scale
Position Error	25	125
Friction		
Room Temperature	15	150
High Temperature	15	150
Low Temperature	20	200
Vibration		
Pointer Oscillation	50	250
Pointer Variation	50	250
Upscale-downscale reading variation	25	200
Voltage & Frequency Variation System Basis	1/2% of point above 1500 PPH except break point and 7.5 PPH below 1500 PPH.	

TABLE III

Flow Rate PPH	Maximum Allowable Pressure Drop PSI
6000	0.21
12000	0.81
15000	1.3

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4.2 Test methods. -

4.2.1 Sampling plan A. -

4.2.1.1 Pressure. - The pressure drops of the transmitter shall be determined for the rates of flow specified and shall not exceed the tolerance specified in Table III.

4.2.1.2 Transmitter vibration during pointer oscillation and vibration tests. - The pointer oscillation and variation test errors shall be determined when the transmitter is subjected to the frequency range at vibratory accelerations or double amplitudes shown below:

Frequency	Amplitude
5 - 100 cps	0.010 inch double amplitude
100 - 500 cps	5 g acceleration

4.2.2 Sampling plan B. -

4.2.2.1 Altitude Test. - This test shall not be required.

4.2.2.2 Low temperature operation. - The transmitter shall be subjected to a temperature of -65° ±2°C for a period of not less than 24 hours, and then operated at this temperature over the scale, temperature shall then be raised to -55° ±2°C for a period of not less than 24 hours. The scale errors and friction shall not exceed the low temperature tolerances as shown in Tables I and II. The transmitter shall be subjected to the room temperature scale error test not less than 1 hour following the cold exposure test. The room temperature tolerances as specified in Tables I and II shall apply.

4.2.2.3 High temperature operation. - The transmitter shall be subjected to an ambient temperature of 121°C for a period of 4 hours with fuel in the instrument, but not flowing. During the last 30 minutes of this period, fuel at 38° ±2°C shall be pumped through the transmitter at a rate of 1,000 pounds per hour. The transmitter shall then be tested for scale errors and friction under the same conditions. The scale errors and friction shall not exceed the tolerances specified in Tables I and II.

4.2.2.4 High temperature exposure. - The transmitter shall be subjected to a temperature of +150° +2°C for a period of 1 hour. The transmitter shall be in the inoperative condition with no fuel flowing through the instrument. At the end of the 1 hour

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exposure the transmitter shall be maintained at room temperature for not less than 4 hours. At the end of the 4 hour period the transmitter shall be tested for room temperature scale errors and friction. The tolerances as specified in Tables I and II shall apply.

4.2.2.5 Endurance. - The scale errors after endurance test shall not exceed the tolerances specified in Table I for the room temperature scale error test by more than 40 pph in the lower scale range, and 100 pph in the upper range. Friction errors shall be within the tolerances specified in Table II.

4.2.2.5.1 Endurance duration. - In the event the elapsed time of the endurance test is less than 1200 hours when the specified number of flow applications has been completed, then the system shall be operated at 80 ±5 percent of the range of the system until a total of 1200 hours of operation has been completed.

4.2.2.6 Contaminated fuel operation. - Test fuel containing dirt to the extent of 80 grams of foreign matter to 1000 gallons of fuel shall be circulated through the transmitter at approximately 2000 pph for one hour. A scale and friction error test shall then be made with the contaminated fuel and operation shall be satisfactory and within the room temperature tolerances. This foreign matter shall be considered to consist of not less than 68 percent SiO_2 and shall have a particle-size analysis as shown in Table IV.

TABLE IV

Foreign Particle Size, Microns	Percent of Total Weight
0 - 5	39 ± 2
5 - 10	18 ± 3
10 - 20	16 ± 3
20 - 40	18 ± 3
over 40	9 ± 3
Through a 200-mesh screen	100 percent

4.2.2.7 Response time. - The fuel flow shall be changed from 0 to 6000 pph by means of a rapid action cut-off valve in the fuel line. The time for the indication of a master indicator connected to the transmitter to come within one percent of its steady state position shall be within 5 seconds.

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4.2.2.8 Shock test. - The test shall be in accordance with Procedure IV Specification MIL-E-5272. The transmitter shall be subjected to impact shocks of 15g, each shock impulse having a time duration 7 ±2 milliseconds. Three shocks shall be applied in each of the following directions:

(a) Along the longitudinal axis
(b) At 45° to the longitudinal axis in one direction.

4.2.2.9 Voltage and frequency variation. - This test shall be conducted as a system test in conjunction with an EFU - 7/A indicator conforming to Specification MIL-I-21206. The applied voltage and frequency for the tests shall be set at the following points. The voltage and frequency test shall be conducted at the 3500 pph and the 12,000 pph flow rates. Readings taken at 115V, 400 cps shall be considered to be reference readings. The readings obtained at the test points above and below normal conditions shall not differ from those obtained at normal conditions by more than the value shown in Table II.

115V	400 cps (Normal conditions)
92V	320 cps
138V	480 cps

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - The transmitter shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method 1A5 omitting preservation compound, using metal reusable containers. (See 6.2).

5.1.1 Level C. - The transmitter shall be packaged in accordance with standard commercial practices.

5.2 Packing. -

5.2.1 Levels A and B. - The transmitter shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract. (See 6.2).

5.2.2 Level C. - The indicator shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers rules and regulation in effect at the time of shipment.

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5.3 Marking. - The interior and exterior containers shall be marked as specified in MIL-P-7936.

5.3.1 Precautionary marking. - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use. - The type fuel flowmeter transmitter covered by this specification is intended to indicate the rate of flow of fuel in pounds per hour being consumed by an aircraft jet engine when used in conjunction with an EFU-7/A indicator conforming to Specification MIL-I-21206 or other suitable indicator.

6.2 Ordering data. - Requisitions, contracts, and orders should state the quantity desired and the levels of packaging (see 5.1) and packing (see 5.2) to be furnished and whether sampling plan B is to be omitted.

6.3 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any

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